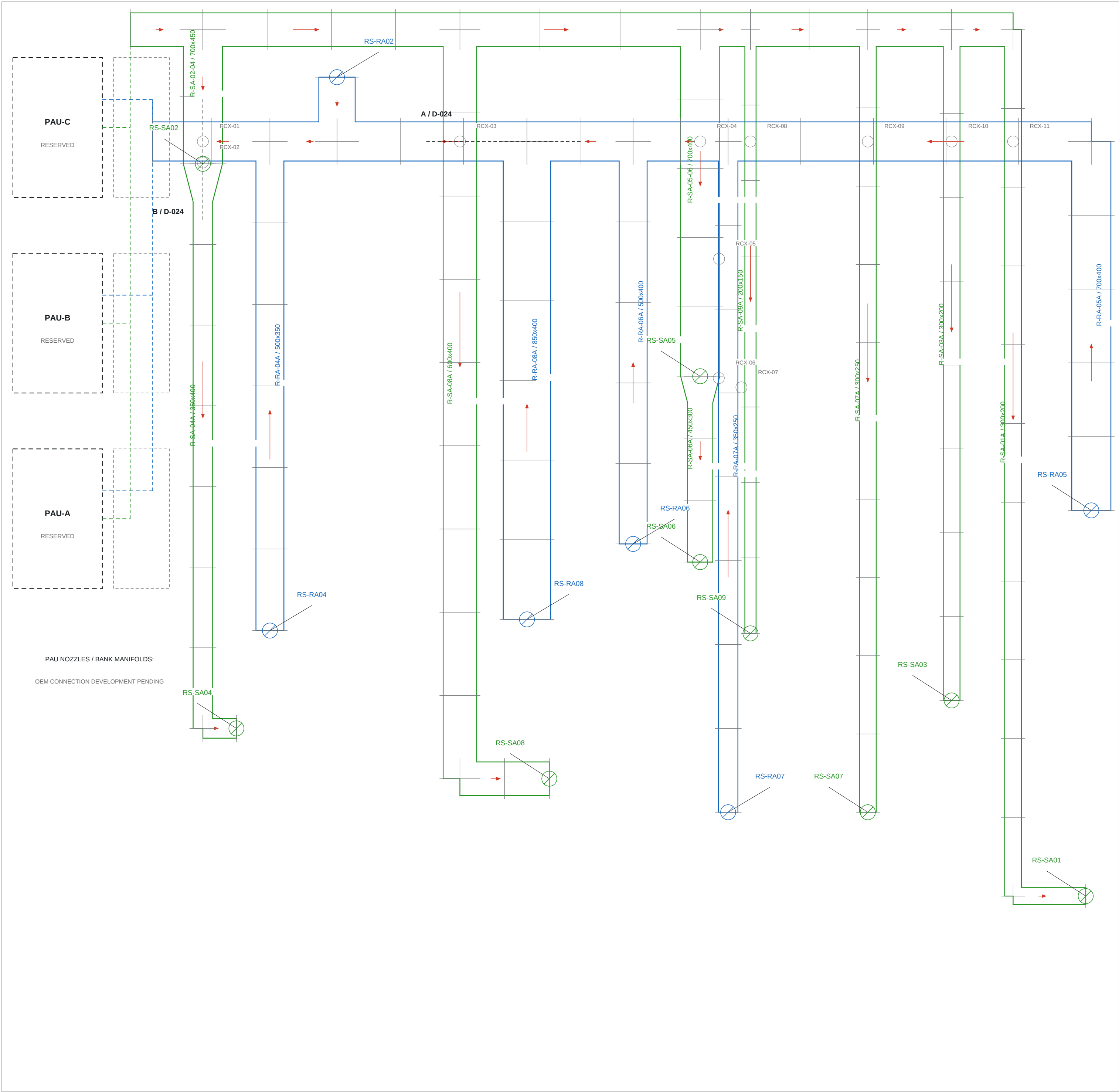




ROOF ROUTES / LEVELS

31 connected route segments feed all 15 room supply / return drops. Coordinates use the supplied floor plan origin. Roof footprint and PAU reservation zones must be surveyed.

GREEN SUPPLY: bare BOD +4.800 m; main 600 x 1300 mm. BLUE RETURN: bare BOD +5.550 m; main 700 x 900 mm. Tall roof mains limit plan width; wind / parapet screening and support stability require structural design.

75 mm outdoor insulation each face assumed. Eleven roof plan overlaps have separated insulation envelopes; minimum calculated separation 150 mm. No upper return riser intersects lower supply insulation in this proposed geometry.

PAU-A / B / C reservations: 1600 x 2500 mm each, 1000 mm east service strip. Dashed nozzle stubs and bank manifolds are schematic interfaces awaiting manufacturer dimensions.

RCX tags: Roof_crossings_Rev12 register. A and B sections: D-024. Curbs and supports: D-030. Native centerline setting-out dimensions: D-032.

Service	Main clear mm	Bare BOD	Insulated top
SA	600 x 1300	+4.800	+6.175
RA	700 x 900	+5.550	+6.525

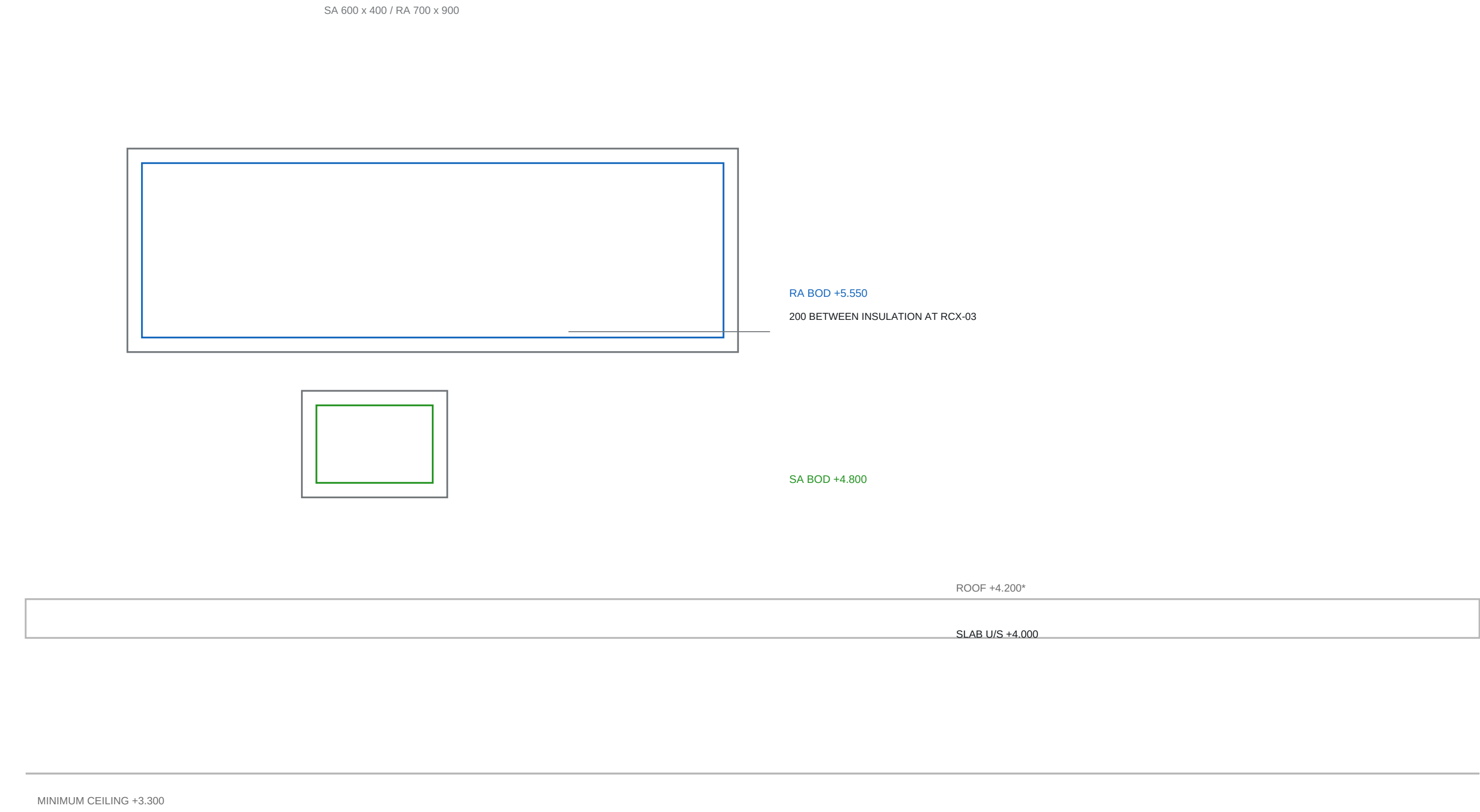
ACCESS / LOAD HOLD

Service strips are reserved for PAUs only. Walking platforms, safe roof access, guarding, filter pull-out, common heater / conditional humidifier weather enclosure, banks and lifting path require selected equipment layout. Roof proposal is not an approved installation.

Roof levels use provisional roof surface +4.200 m: user underside +4.000 m plus assumed 200 mm slab. Route lengths are measured proposals, not surveyed as-built lengths. All sizes are clear internal mm.

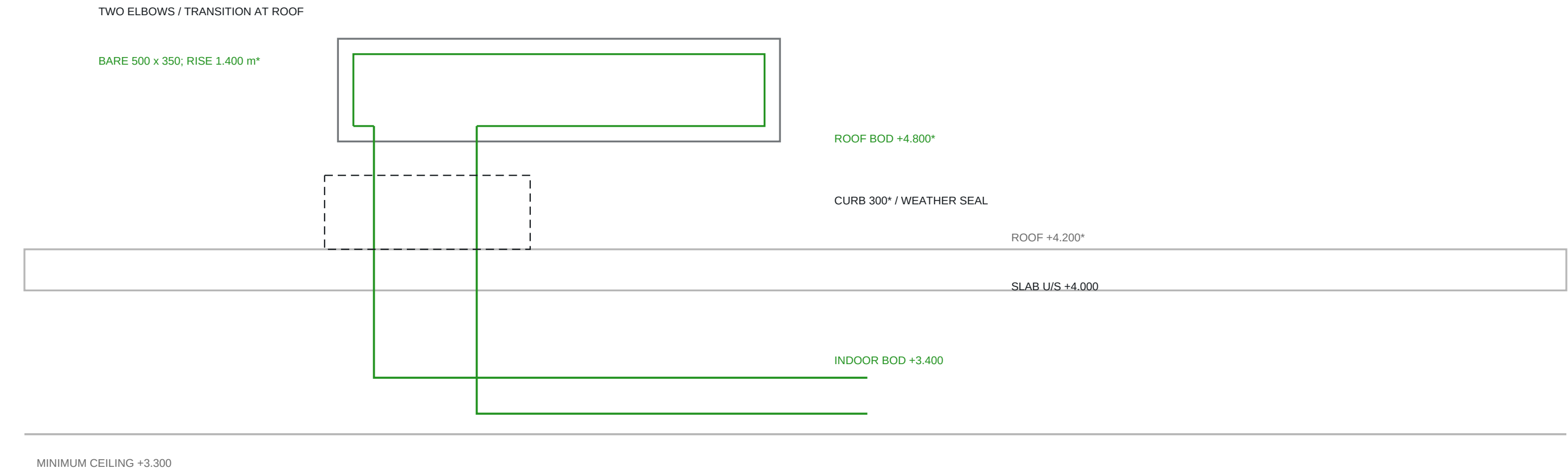
ADMINISTRATION BUILDING - ROOF CROSSING AND ROOM DROP SECTIONS

A RCX-03 / SA08 x RA MAIN | 1:20



* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

B RS-SA02 DROP | VERTICAL 1:20 / HORIZONTAL NTS



* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

VERTICAL DEVELOPMENT / CONNECTION RULES

Riser straight-length schedules use proposed bare BOD differences at equal-size horizontal ends. Elbow development and differing roof header sizes are separate fitting allowances. Maintain 3.300 m finished ceiling; do not lower a room duct to clear a roof route.

Use shop-developed mitred elbows with turning vanes or selected radiused elbows. Square plan junctions on the roof sheet denote fitted tees; they are not four-way crosses. Every room drop has an independent riser ID and reserved curb opening.

Sections are coordination details, 1:20 at A1 where stated; equipment and curb symbols are NTS. Clear opening schedules: Roof_curbts_Rev12. Structural capacity, waterproofing and fire ratings have not been approved.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CROSSING AND ROOM DROP SECTIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-024

SCALE: AS SHOWN AT A1

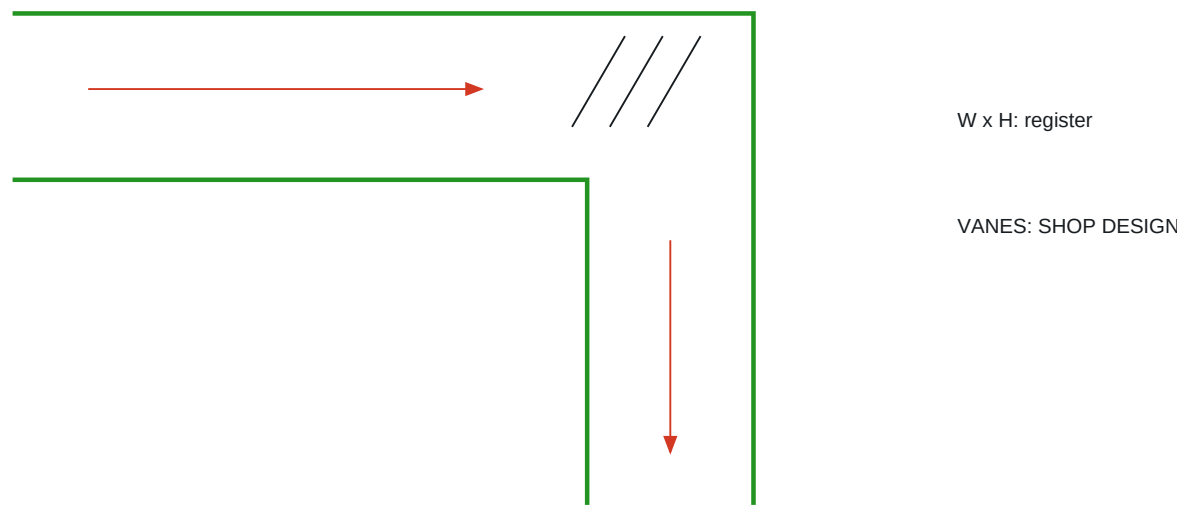
REVISION

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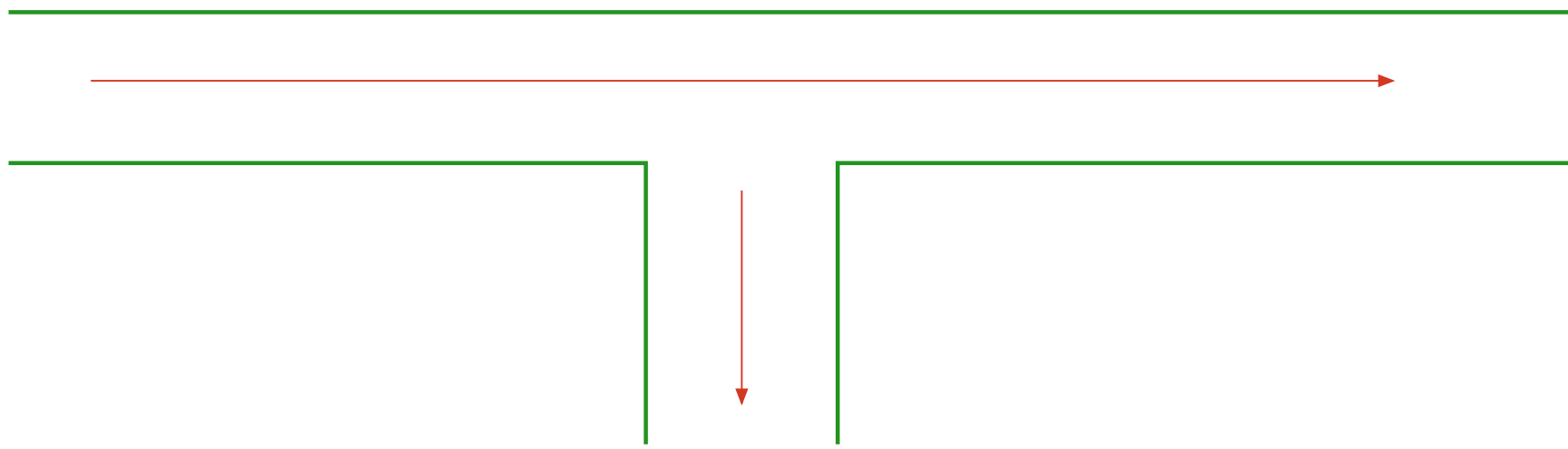
SHEET 24 OF 32

01 MITRED ELBOW / VANES



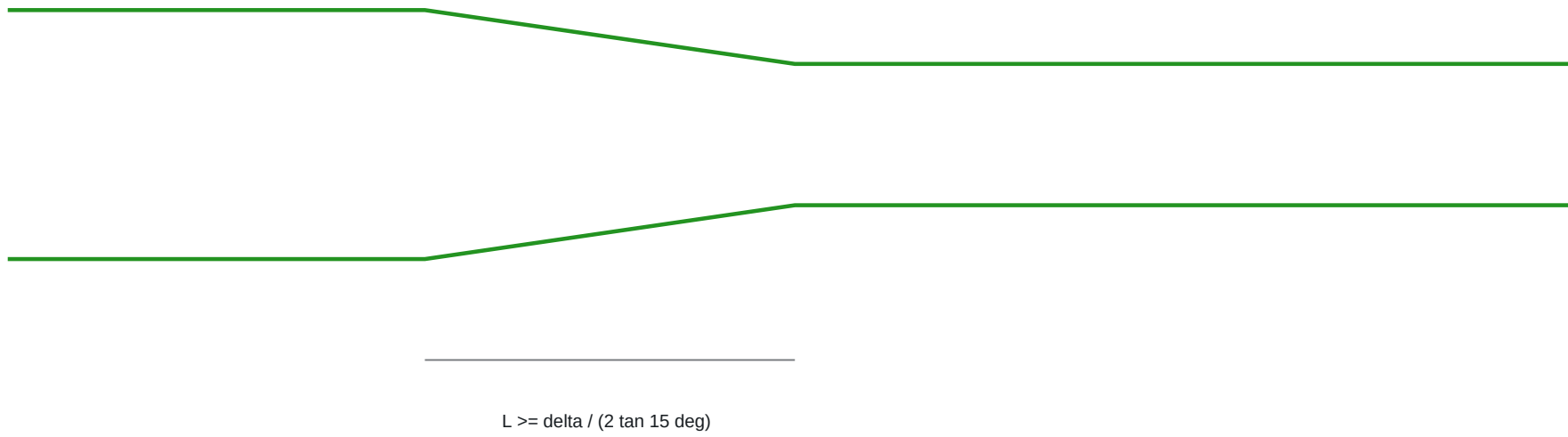
SQUARE THROAT / 45 DEG MITRES

02 THREE-LEG TEE



THREE LEGS ONLY; OPEN BRANCH THROAT

03 NECK / RISER TRANSITION



MAX 30 DEG INCLUDED ANGLE PROPOSED

ALL FITTINGS / DIMENSION SCHEDULE

Fitting_design_Rev12 contains 236 tagged loss / geometry allowances covering indoor path bends, joints, terminal adapters, balancing / conditional fire dampers and roof / riser connections. It is a design development register, not an OEM bill of materials.

For an actual two-dimensional reducer, proposed L = max(300 mm, max(abs(W1-W2), abs(H1-H2)) / (2 tan 15 deg)). Hold the bottom level only if the selected terminal / device height permits; otherwise show the offset in the shop section.

Roof main sizes are constant along each service spine, so no unshown step reductions occur. Branch width / height change at the tee is developed within a shop boot. Keep main and branch walls open at the connected throat.

Bend / tee lengths must fit the reserved straight station before moving any VCD, RCD, MFD, EH or access door. Reference CAD glyph dimensions are symbols; actual face-to-face lengths and actuator envelopes require an OEM selection.

Fitting	Riser	Clear W x H	Elbows / K each
F12-207	RS-SA02	500.0 x 350.0	2 / 0.50 assumed
F12-209	RS-RA02	650.0 x 400.0	2 / 0.50 assumed
F12-211	RS-SA04	350.0 x 400.0	2 / 0.50 assumed
F12-213	RS-RA04	500.0 x 350.0	2 / 0.50 assumed
F12-215	RS-SA08	600.0 x 400.0	2 / 0.50 assumed
F12-217	RS-RA08	850.0 x 400.0	2 / 0.50 assumed
F12-219	RS-SA05	600.0 x 350.0	2 / 0.50 assumed
F12-221	RS-RA05	700.0 x 400.0	2 / 0.50 assumed
F12-223	RS-SA06	450.0 x 300.0	2 / 0.50 assumed
F12-225	RS-RA06	500.0 x 400.0	2 / 0.50 assumed
F12-227	RS-SA09	200.0 x 150.0	2 / 0.50 assumed
F12-229	RS-SA07	300.0 x 250.0	2 / 0.50 assumed
F12-231	RS-RA07	350.0 x 250.0	2 / 0.50 assumed
F12-233	RS-SA03	300.0 x 200.0	2 / 0.50 assumed
F12-235	RS-SA01	300.0 x 200.0	2 / 0.50 assumed

ACTUAL ROOF BRANCH TAPERS - mm

Route	Inlet	Outlet	Developed L
R-SA-04A	700x450	350.0x400.0	675
R-SA-06A	700x400	450.0x300.0	475

All K values on this issue are explicit preliminary allowances, not selected catalogue coefficients. Adapter_details_Rev12 retains the actual connected terminal adaptor coordinates / proposed lengths. Fabrication approval remains open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FITTING DEVELOPMENT AND FABRICATION RULES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-025

SCALE: NTS AT A1

REVISION

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ADMINISTRATION BUILDING - SYSTEM PRESSURE LOSS AND FAN DUTY BUDGET

System	Duty each L/s	Estimated external Pa	Budget Pa +20%	Low / high sensitivity Pa
PAU bank - combined external SA + RA path	1848.90	329.9	400	169.7 / 490.1
EF-03 duty / standby	147.60	30.7	40	15.7 / 45.7
EF-09 duty / standby	64.89	33.6	50	17.6 / 49.6

CRITICAL TERMINAL PATHS - ALL 52 IN REGISTER

Terminal	Air	Straight Pa	Fittings Pa*	Heater Pa*	Common Pa*	Terminal Pa*	Total Pa*
SD-05-01	SA	5.802	55.411	25	150	30	266.212
SD-05-02	SA	5.728	55.411	25	150	30	266.138
SD-05-03	SA	5.827	55.147	25	150	30	265.974
SD-05-04	SA	5.347	54.555	25	150	30	264.901
SD-05-05	SA	5.273	54.555	25	150	30	264.828
SD-06-02	SA	5.962	43.929	25	150	30	254.892
SD-04-02	SA	5.911	43.476	25	150	30	254.387
SD-06-03	SA	6.124	42.891	25	150	30	254.014
SD-04-03	SA	6.005	42.646	25	150	30	253.651
SD-06-01	SA	5.939	40.843	25	150	30	251.782
SD-04-01	SA	5.514	40.284	25	150	30	250.798
SD-08-06	SA	7.099	54.359	0	150	30	241.458

METHOD / BUDGET LIMITS

Straight friction: Darcy-Weisbach / Colebrook, rho 1.20 kg/m3, mu 1.81e-5 Pa.s, galvanized roughness 0.09 mm, hydraulic diameter 2WH/(W+H). Loss = L x friction + sum(K rho V2/2) + component allowance.

Assumed K: elbows 0.50; tee branch 1.00; tee straight run 0.20; joint bend 0.50; adapter 0.25; balancing damper 2.00; conditional fire damper 1.00. Terminal 30 Pa, room heater 25 Pa, supply common external equipment 150 Pa. Sensitivity scales all assumed local / equipment losses to 50% and 150%.

PAU budget adds worst SA and RA terminal paths. Unit internal coils / dirty filters belong in the OEM fan calculation with explicit external-pressure convention. PAU nozzle / bank manifolds, OA intake, selected dampers, silencers and exhaust discharge are not measured in this model; no fan is selected.

Pressure paths include measured proposed roof plan segments and provisional riser rises, plus retained indoor centerlines. Budget values are screening estimates; fan curves, dirty-filter losses and rotating-duty bank cases require OEM confirmation.

TAQA TRANSMISSION N-19082-RO-012 PUMPING STATION STANDARDIZATION ADMINISTRATION HVAC DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE	DRAWING TITLE SYSTEM PRESSURE LOSS AND FAN DUTY BUDGET FOR COORDINATION CHECKED / APPROVED: PENDING	DRAWING NUMBER ADM-HVAC-D-026 SCALE: NTS AT A1	REVISION 12 08 OCT 2026 SHEET 26 OF 32
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ADMINISTRATION BUILDING - AIR BALANCE AND TERMINAL SELECTION CRITERIA

GF	Supply L/s	Return L/s	Known extract L/s	Net transfer / leakage L/s	Pressure target / verification
02	664.80	605.46	0.00	59.34	50 Pa / not verified
04	449.20	389.86	0.00	59.34	50 Pa / not verified
08	884.60	825.26	0.00	59.34	50 Pa / not verified
05	777.70	718.36	0.00	59.34	50 Pa / not verified
06	418.20	358.86	0.00	59.34	50 Pa / not verified
09	46.20	0.00	64.89	-18.69	25 Pa / not verified
07	168.20	108.86	0.00	59.34	50 Pa / not verified
03	147.60	0.00	147.60	0.00	25 Pa / not verified
01	141.30	0.00	0.00	141.30	25 Pa / not verified

OUTDOOR AIR / PRESSURE ACTION

Mass-balance minimum OA at the retained SA / RA duty is 691.12 L/s, before any code or process ventilation requirement. Current OA reference 420.70 L/s is short by 270.42 L/s. This is a required design reconciliation, not a selected OA duty.

Known extract is 212.49 L/s; residual exfiltration / transfer is 478.63 L/s less the unresolved clean-agent extract. Verify each door / transfer path and exterior leakage before asserting +50 / +25 Pa.

Toilets have -18.69 L/s net supply minus extract; kitchen has zero net. Positive pressure targets need owner / process / hygiene reconciliation and a measured leakage basis. Do not reduce extract simply to obtain a pressure label.

SUPPLY NECK / COVERAGE SCREEN - 53 ROW REGISTER

Representative	L/s	Neck	m/s	Half spacing m*
SD-02-01	166.20	350	1.727	2.2
SD-05-01	155.54	325	1.875	2.25
SD-04-01	149.73	325	1.805	1.626
SD-08-01	147.43	325	1.777	1.15
SD-06-01	139.40	300	1.972	4.65
SD-07-01	168.20	350	1.748	Room coverage required
SD-09-01	23.10	150	1.307	0.9
SD-03-01	147.60	325	1.779	Room coverage required
SD-01-01	141.30	300	1.999	Room coverage required

* Half spacing is a coverage screening distance, not certified throw. Require OEM pressure loss, NC and isothermal / cooling throw at 0.25 m/s terminal velocity; check wall coverage, induction, ADPI and occupied-zone draught.

Proposed noise brief. NC <=30 meeting / rest / operator rooms; NC <=35 other occupied rooms, subject to owner / acoustic confirmation. No catalogue model, throw or NC has been invented.

Duty, neck velocities and same-type spacing are calculated from scheduled terminal coordinates. Air_balance_Rev12 and Terminal_performance_Rev12 are editable calculation / selection registers; existing duties are retained pending engineering reconciliation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

AIR BALANCE AND TERMINAL SELECTION CRITERIA

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-027

SCALE: NTS AT A1

REVISION

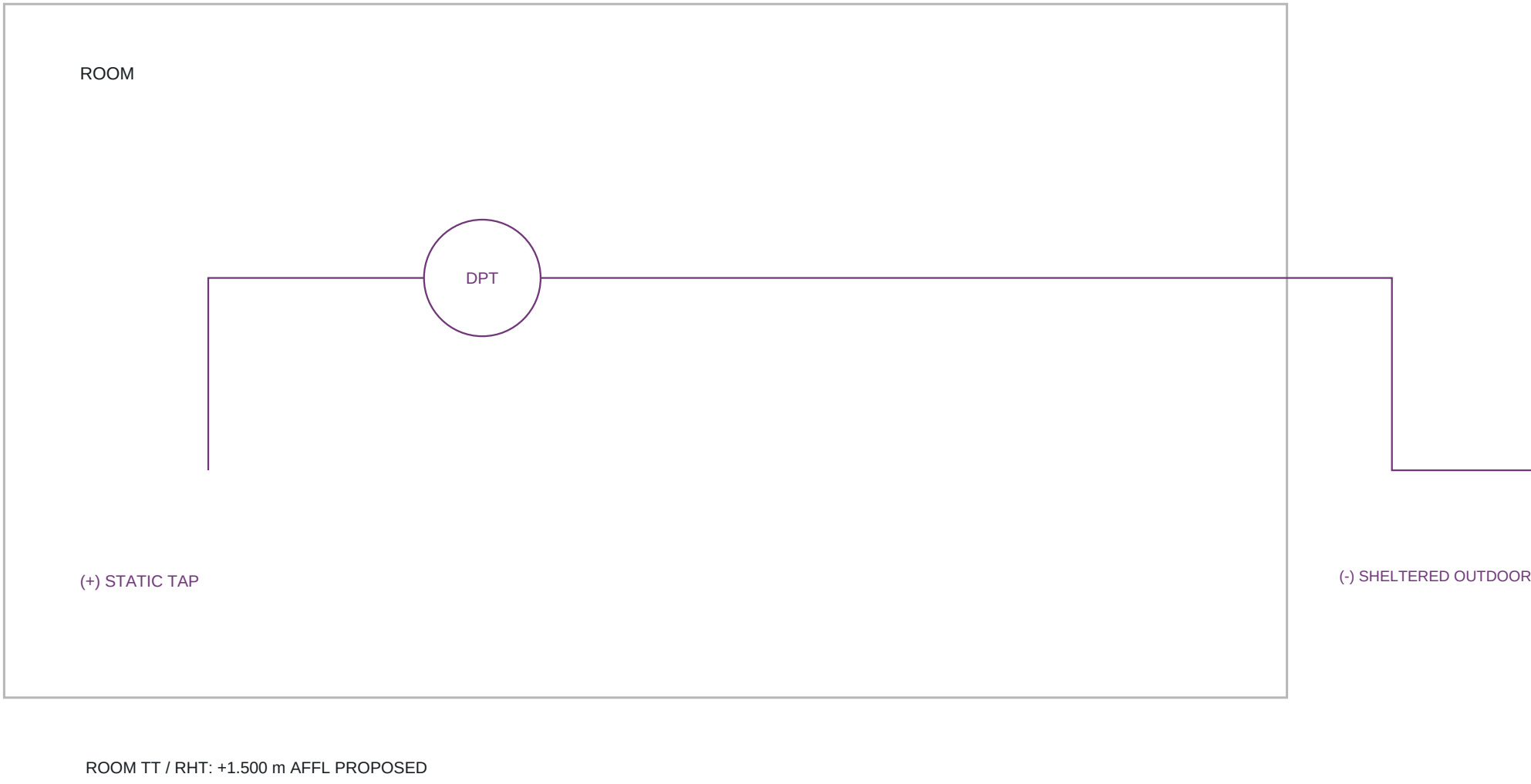
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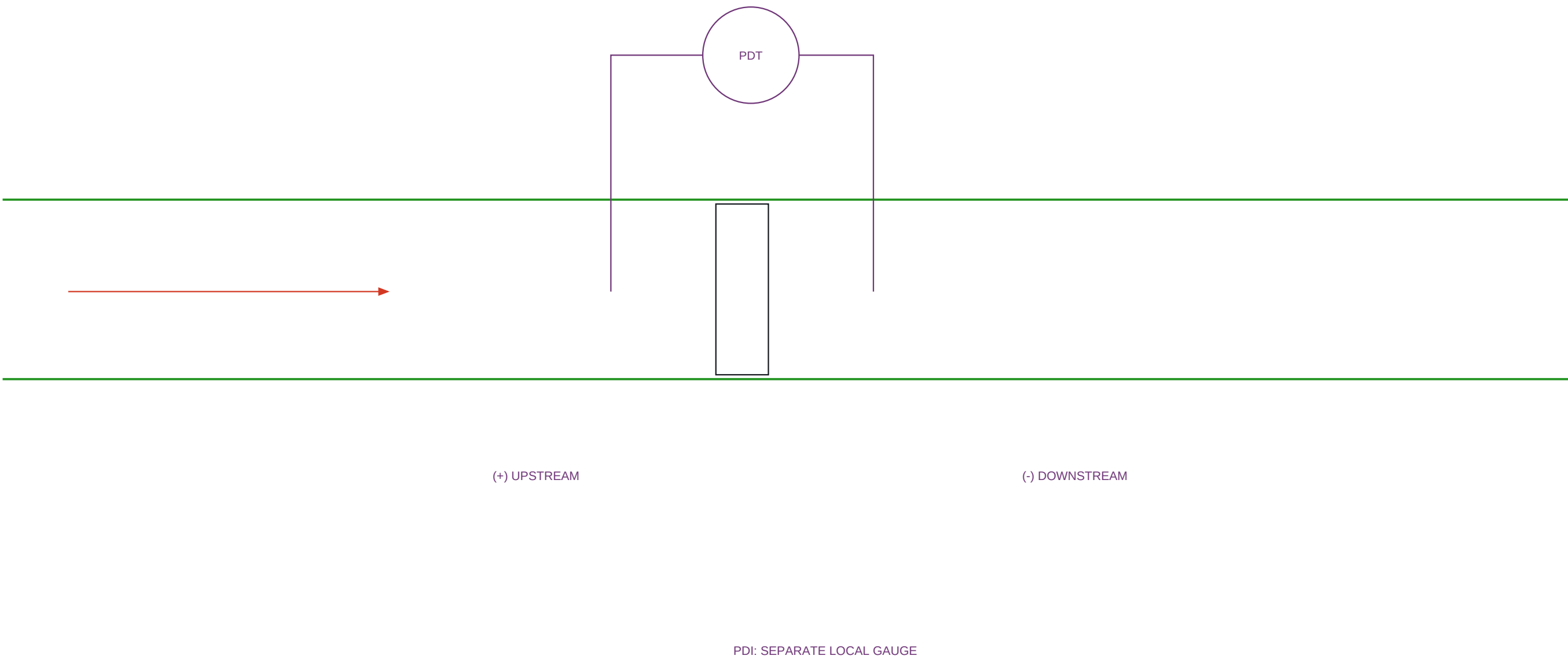
SHEET 27 OF 32

ADMINISTRATION BUILDING - INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

A ROOM STATIC PRESSURE



B FILTER / FAN DIFFERENTIAL



INSTALLATION SPECIFICATION - ALL 85 TAGS

Instrument_installation_Rev12 gives mounting, provisional ranges, pressure connections, wiring, access and commissioning requirements for every instrument. Tag requirement and I/O duties are retained from the approved input register, not replaced with generic symbols.

Proposed room DPT range -100..+100 Pa; fan / filter differential 0..1000 Pa; TT 0..50 degC; RH 0..100%; vibration 0..25 mm/s RMS. Ranges are selection proposals, not final alarm / trip settings.

Use labelled static pressure pickups and equal reference datums. Protect outdoor reference from wind / rain; avoid jet impingement, condensate traps and unserviceable tubing. Provide isolation and test ports per manufacturer.

COMMON-DUCT AIRFLOW / SENSOR ACCESS

FIT-SA-01 / FIT-RA-01: reserve 5Dh upstream and 3Dh downstream as a screening space allowance. Actual manufacturer measurement station / traversing method governs; no accuracy claim at the provisional roof position.

Main SA Dh = 0.821 m: screen 4.11 m upstream / 2.46 m downstream. Main RA Dh = 0.788 m: screen 3.94 m upstream / 2.36 m downstream. The current short bank-to-first-branch space does not prove these lengths; measurement station integration is an open hold.

Room temperature / humidity sensors avoid direct supply jets, exterior solar gain and heaters. VS installs on approved bearing / frame position. TSHH and AFS remain accessible for test / manual reset; selected heaters govern technology and settings.

Installation detail references D-028 / D-029 supplement D-012 to D-016 and the 85-row index. No final cable route, panel terminal number, approved sensor accuracy or OEM setpoint is asserted.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-028

SCALE: NTS AT A1

REVISION

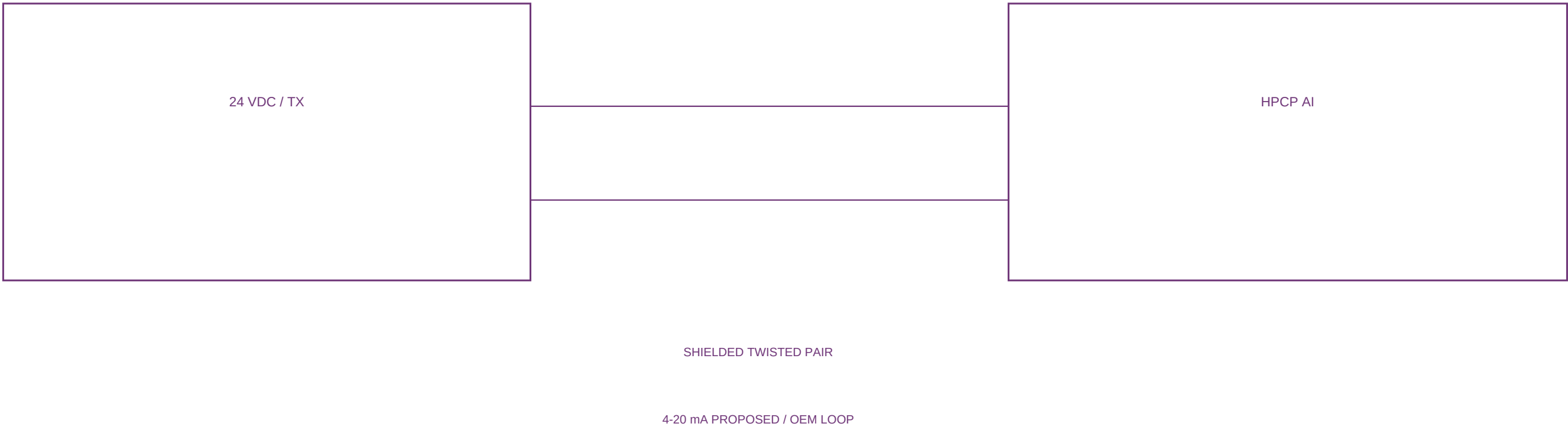
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08 OCT 2026

SHEET 28 OF 32

ADMINISTRATION BUILDING - I&C WIRING AND HEATER SAFETY INTERFACES

A ANALOG TRANSMITTER / AI



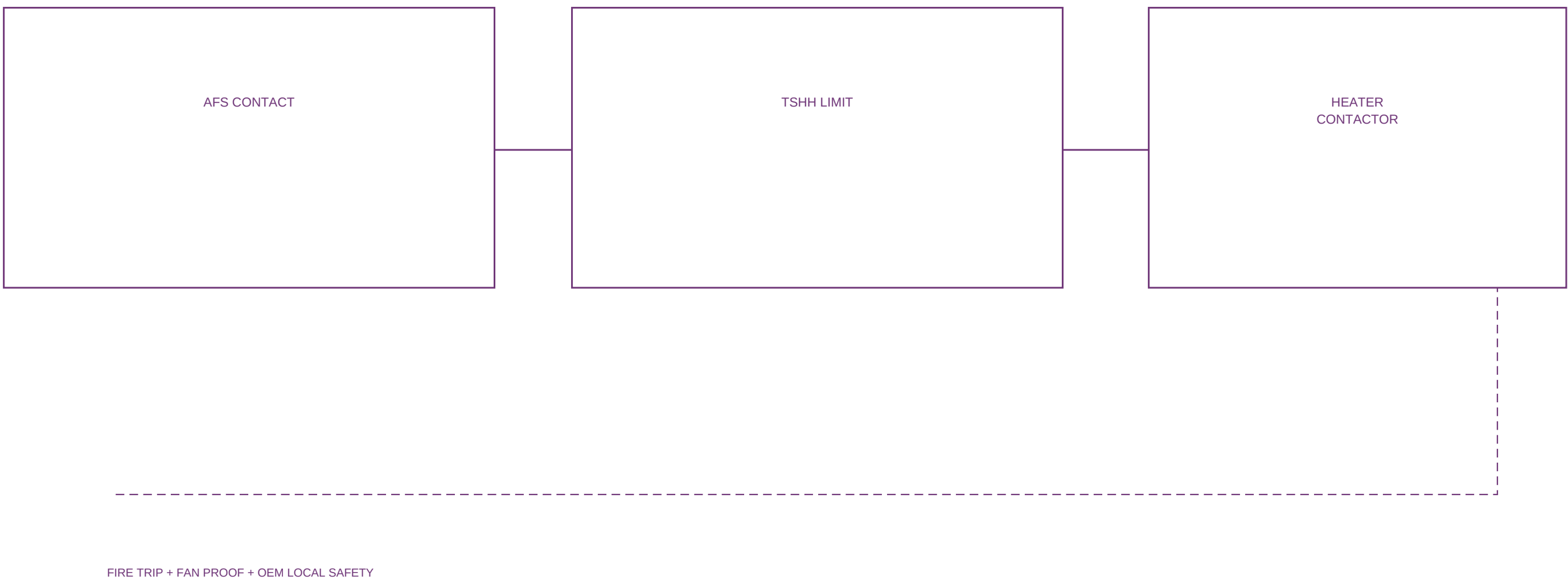
SIGNAL / PANEL REQUIREMENTS

AI proposals use isolated 4-20 mA channels and shielded twisted pairs. Confirm transmitter supply, two / three / four wire circuit, loop burden and earthing with OEM. Earth the cable shield at the panel end per project EMC design; segregate from mains / VFD cables.

DI proposals are dry contacts. Contact state, wetting voltage, galvanic isolation and fail-state indication require manufacturer wiring. Local PDI is not an AI. COMM energy meter protocol, CT ratio, address and termination remain pending.

Every tag maps to the unchanged 275-row I/O register. Final channel numbers, junction box locations and cable lengths await panel design and coordinated routes. Commission with traceable point-to-point checks and calibrated simulated signals.

B HARDWIRED HEATER PERMISSIVES



SAFETY / CAUSE AND EFFECT

Six AFS and six independent TSHH heater provisions remain present. Loss of airflow or high-limit trip inhibits heat through the OEM hardwired circuit. PLC heating demand cannot bypass either permissive. Alarm feedback reports to HPCC.

Confirmed fire stops all HVAC fans and heaters. No automatic fire extraction / purge is added. Fire damper rating, fail position, reset procedure and boundary quantity follow approved fire strategy. Manual reset is required after confirmed fire.

This is an interface diagram, not an electrical terminal drawing. Complete the heater OEM wiring, upstream electrical protection, safety relay architecture and fire interface review before procurement.

Input requirement and I/O assignments are retained. Final wiring and approved safety settings require selected hardware. Instrument_Installation_Rev12 is the 85-row specification companion.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

I&C WIRING AND HEATER SAFETY INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-029

SCALE: NTS AT A1

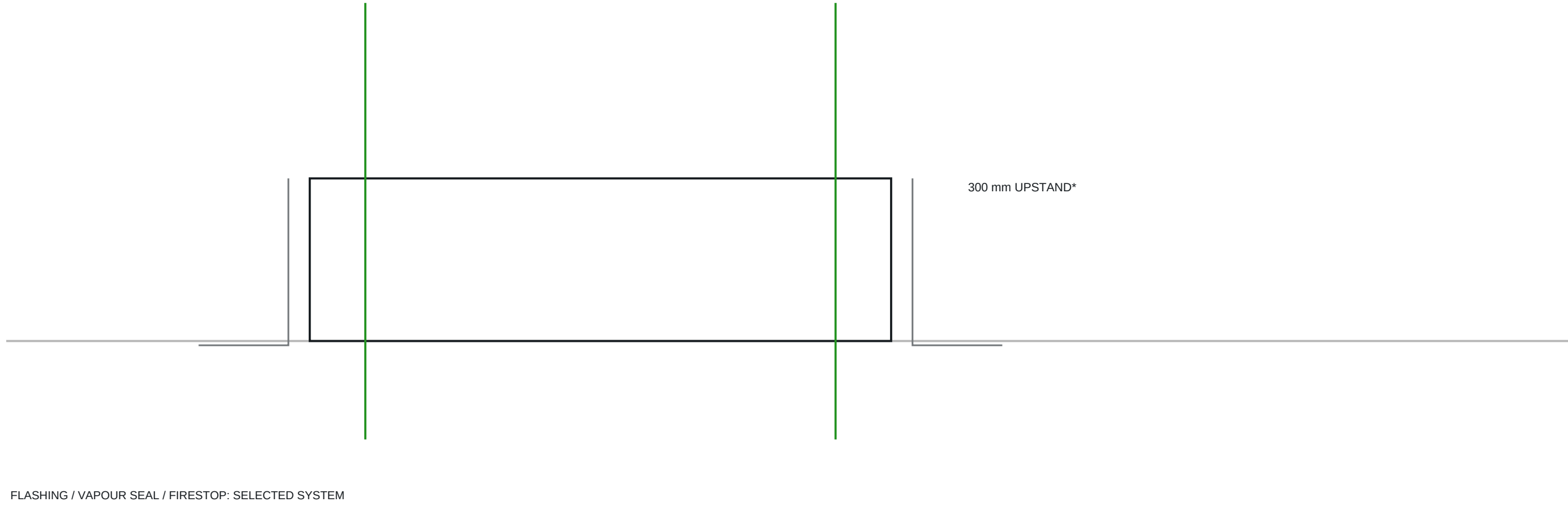
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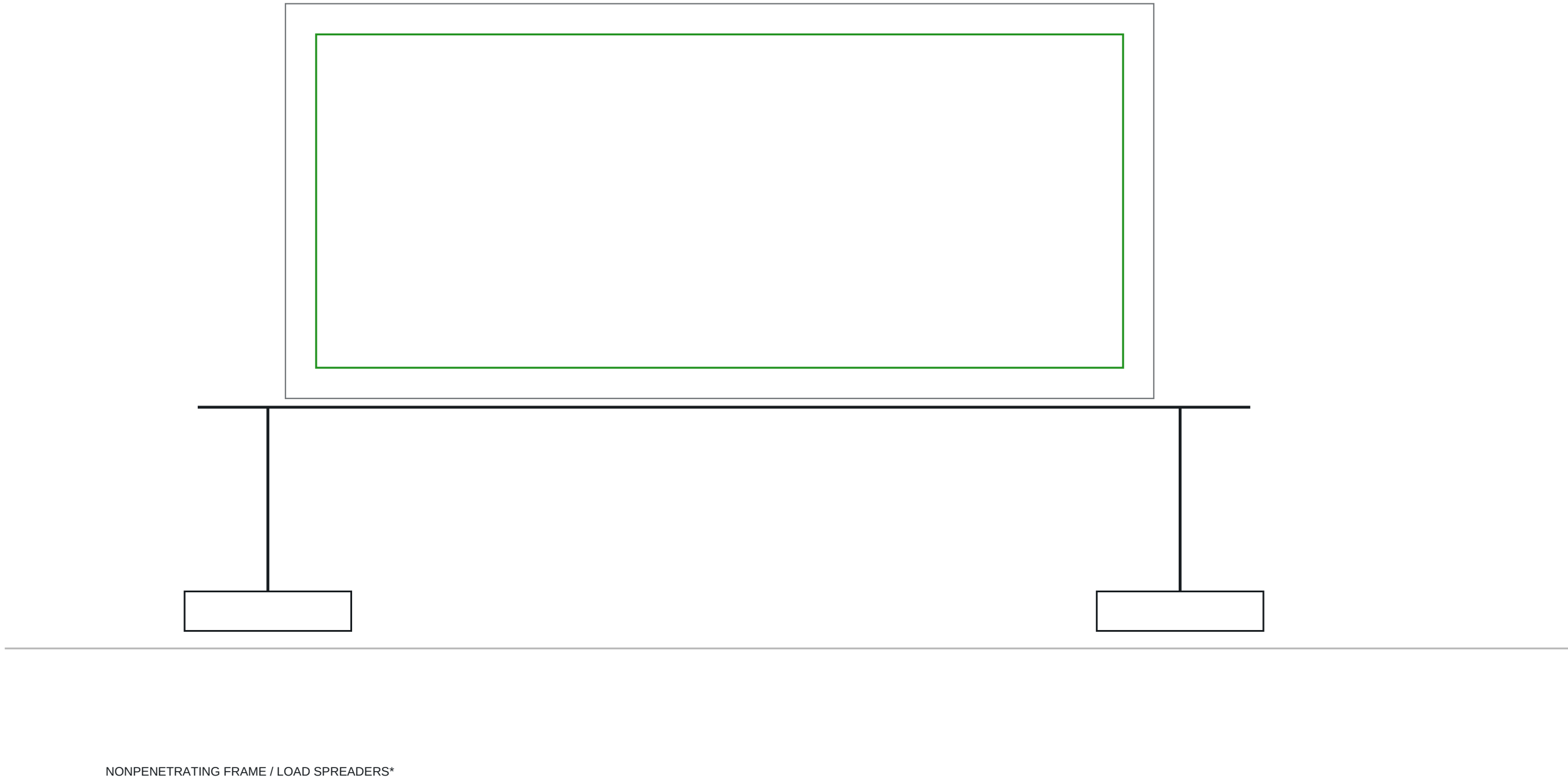
A PROPOSED CURB SECTION - NTS



CURBS / PENETRATIONS

- 15 curb reservations have X / Y coordinates and opening envelopes: clear duct plus 75 mm insulation and 100 mm installation allowance each side. Upstand 300 mm above proposed roof surface; confirm finished waterproofing datum and local exposure.
- Coordinate opening clearances, sleeves, firestop and insulated riser separation with structural and fire disciplines. Keep weatherproofing continuous; selected flashing / vapour seal and cladding joints must shed rain. Do not cut a slab or beam from this drawing.
- Outdoor insulation 75 mm is provisional. Specify thermal conductivity, vapour barrier, sealed cladding, weather / UV resistance and fire classification from approved project materials. It is not a declared UAE average.

B ADJUSTABLE SUPPORT FRAME - NTS



SUPPORT / ACCESS PROCUREMENT HOLD

- 157 support station reservations, maximum nominal 1500 mm route spacing. Deduplicate junction stations and adjust fitting / equipment supports in shop drawings. Geometry alone does not select a bracket or roof anchor.
- Support load table remains TBC until gauge, fitting weight, insulation / cladding, actuator, wind, seismic requirements and frame self-weight are known. Tall roof mains need wind stability / bracing and maintenance platform design.
- Verify structure and membrane bearing / protection before choosing nonpenetrating ballast or anchors. PAU lifting, coil / filter withdrawal and heater / humidifier service envelopes require a complete manufacturer footprint.

* Proposal only, Roof_curbs_Rev12 and Roof_supports_Rev12 provide setting-out reservations, not anchor design or certified structural loads. Roof surface +4.200 m assumes 200 mm structural slab above the confirmed +4.000 m underside.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CURB, SUPPORT AND WEATHERPROOFING

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-030

SCALE: NTS AT A1

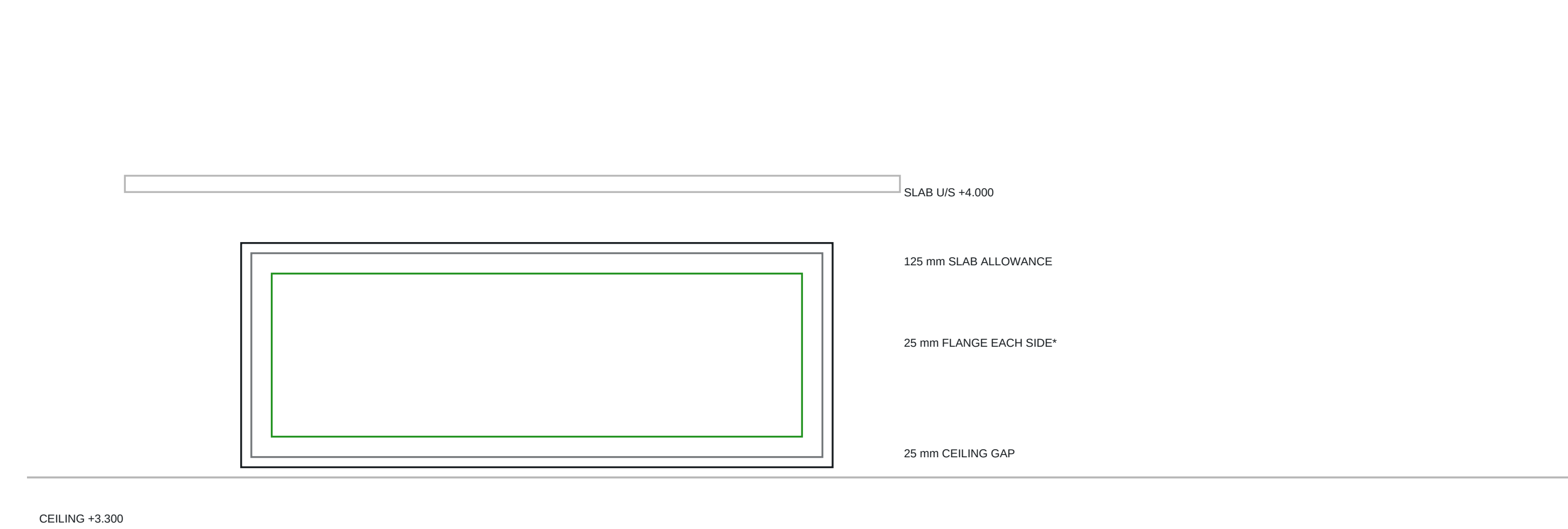
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A DEEPEST DUCT / FLANGE | 1:10



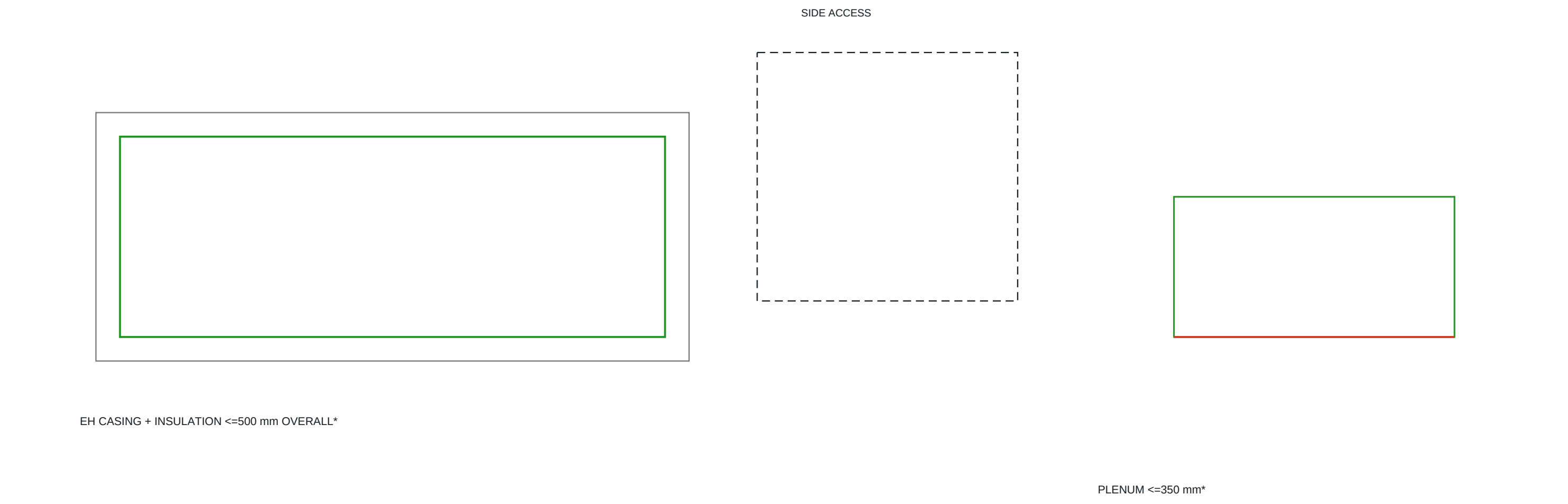
DUCT + FLANGES

Deepest bare depth 400 mm, BOD +3.400 m, 50 mm insulation each face, proposed 25 mm flange allowance beyond insulation: overall reserved depth 550 mm. Lowest +3.325 m; highest +3.875 m. Ceiling gap reduces to 25 mm; slab allowance reduces to 125 mm.

This is a fit screen, not a maintenance clearance. Hanger / beam / other-service and flange shape / vapour seal require shop coordination. An actuator above the duct can exceed this envelope: select a side actuator / lower-profile assembly or locally reroute.

Installation_envelopes_Rev12 checks every indoor duct and specifies room-heater procurement limits. No selected component envelope is represented as verified.

B ROOM HEATER / PLENUM | NTS



HEATER / TERMINAL PROCUREMENT LIMITS

Room heater casing plus insulation must fit between +3.350 and +3.850 m (500 mm overall) unless separately coordinated. Reserve side service access and an aligned ceiling hatch for reset / removal. OEM isolation and thermal clearances govern.

A 350 mm overall terminal plenum above ceiling +3.300 m reaches +3.650 m. Provide an offset / vertical neck to the branch; verify the actual plenum, collar and flex radius in the 700 mm ceiling void. A 600 mm face is not a 600 mm-deep plenum.

All access doors and actuators need a removal path. Final hatch dimensions, heater face-to-face length and damper / actuator position cannot be approved from symbolic plan blocks. If OEM envelopes exceed limits, revise the local branch before release.

* Explicit coordination / procurement assumptions, not selected equipment dimensions. A gives a vertical envelope at 1:10 at A1; component illustrations in B are NTS. Minimum clear height remains +3.300 m.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

CEILING COMPONENT AND ACCESS ENVELOPES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-031

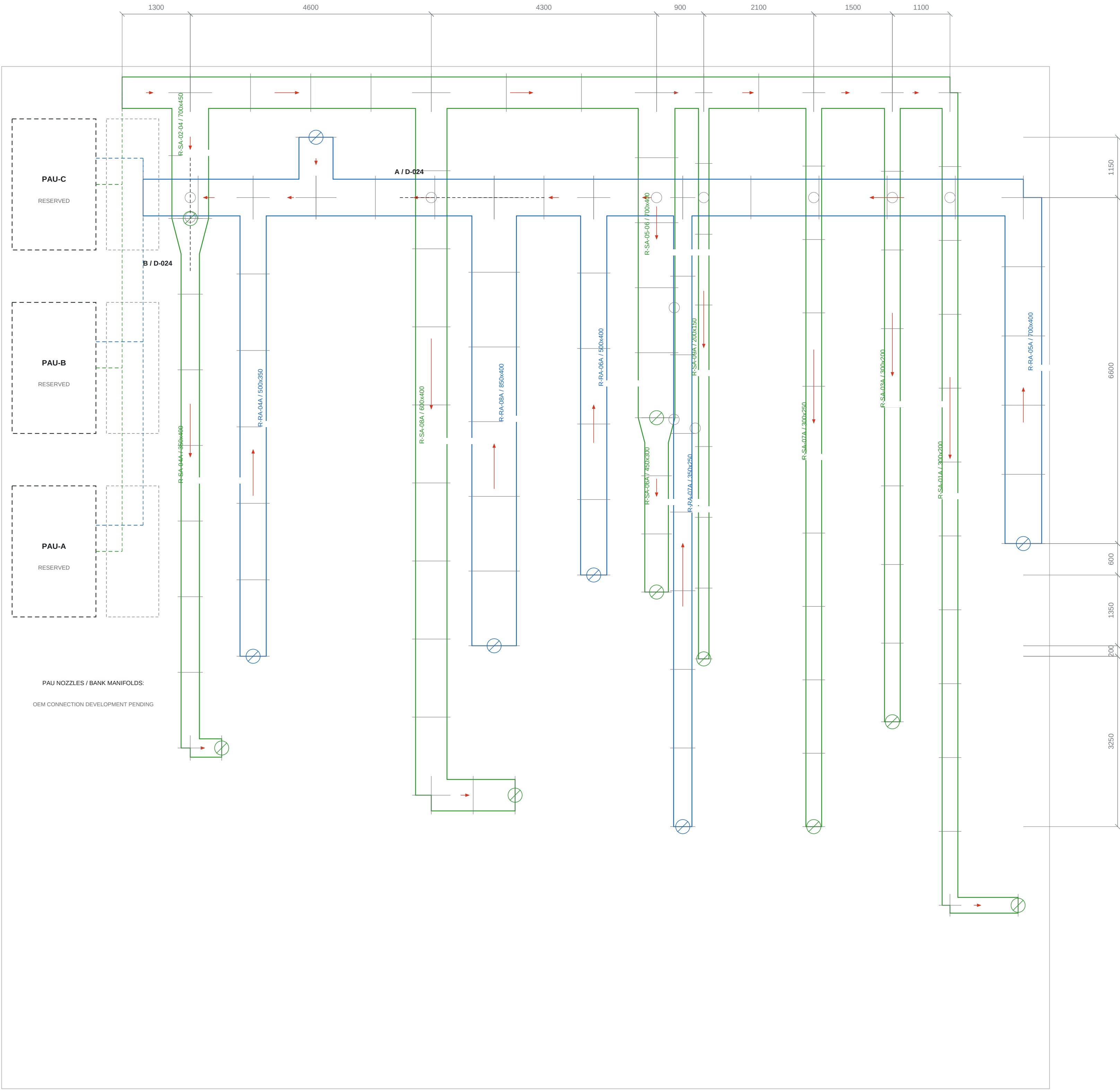
SCALE: AS SHOWN AT A1

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SHEET 31 OF 32



PROPOSED ROOF DROP COORDINATES - mm

Riser	X	Y	Clear mm	Rise m*
RS-SA02	3600	16600	500 x 350	1.4
RS-RA02	6000	18150	650 x 400	2.15
RS-SA04	4200	6500	350 x 400	1.4
RS-RA04	4800	8250	500 x 350	2.15
RS-SA08	9800	5600	600 x 400	1.4
RS-RA08	9400	8450	850 x 400	2.15
RS-SA05	12500	12800	600 x 350	1.4
RS-RA05	19500	10400	700 x 400	2.15
RS-SA06	12500	9475	450 x 300	1.4
RS-RA06	11300	9800	500 x 400	2.15
RS-SA09	13400	8200	200 x 150	1.4
RS-SA07	15500	5000	300 x 250	1.4
RS-RA07	13000	5000	350 x 250	2.15
RS-SA03	17000	7000	300 x 200	1.4
RS-SA01	19400	3500	300 x 200	1.4

SETTING-OUT / REGISTER LINKS

13 native roof dimension segments supplement the 11 retained architectural dimensions. Individual route end coordinates and lengths are in Roof_routes_Rev12; curbs use the same XY origin.

Local origin is the supplied floor drawing (0,0), not a surveyed benchmark. Verify parapets, equipment plinths, openings, wind screening, beam grids and vertical datums before establishing roof setting-out.

* Rises use provisional roof levels; bend / reducer development and selected PAU nozzle connections are separate. Bank interface stubs are schematic, not dimensioned equipment outlets.

Source footprint 20000 x 19500 mm. Dimensions indicate proposed route centerline coordinates. Nominal units mm / m AFFL; proposed heights and fabrication details require professional review and coordinated OEM submissions.